

Lecture 6 - Part II - Algebra of Taxes and Subsidies

ECON 101
Fall 2020

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Demand: $Q_D = Q_D(P^d)$; P^d price paid by buyers

Supply: $Q_S = Q_S(P^s)$; P^s price received by sellers

In the absence of any restriction:

Market equilibrium: $P^E = P^d = P^s$

$$Q_D(P^E) = Q_S(P^E) \Rightarrow Q^E$$

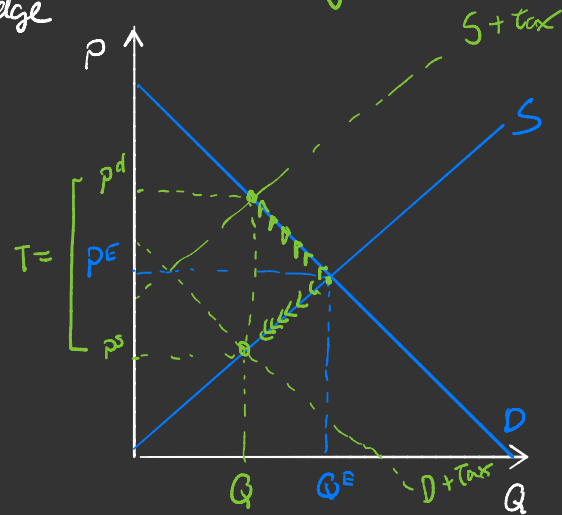
When there is a per-unit tax: introduces a wedge between P^d and $P^s \Rightarrow P^d \neq P^s$ in fact

we will have $T = P^d - P^s$ this, together with

$$Q_D(P^d) = Q_S(P^s)$$

Form a system of equations in which we need to find P^d, P^s, Q

These shifts
 $S+Tax$ OR $D+Tax$
are auxiliary in
understanding what's
happening!



Assuming Linear demand and supply:

$$Q_D = a - bP^D, \quad Q_S = c + dP^S, \quad a, b, c, d > 0$$

The tax-per unit $T = P^D - P^S$
traded

① Therefore: $P^S = P^D - T \Rightarrow$ "Auxiliary shift in supply"

② We could have $P^D = T + P^S \Rightarrow$ "Auxiliary shift in demand"

① $Q_S = c + d(P^D - T)$; ② $Q_D = a - b(T + P^S)$

↓
To obtain P^D , P^S and Q we solve:

$$a - bP^D = c + d(P^D - T) \Rightarrow$$

$$P^D = \frac{a - c + dT}{b + d}$$

$$P^S = P^D - T = \frac{a - c + dT}{b + d} - T = \frac{a - c + dT - bT - dT}{b + d}$$

$$P^S = \frac{a - c - bT}{b + d}$$

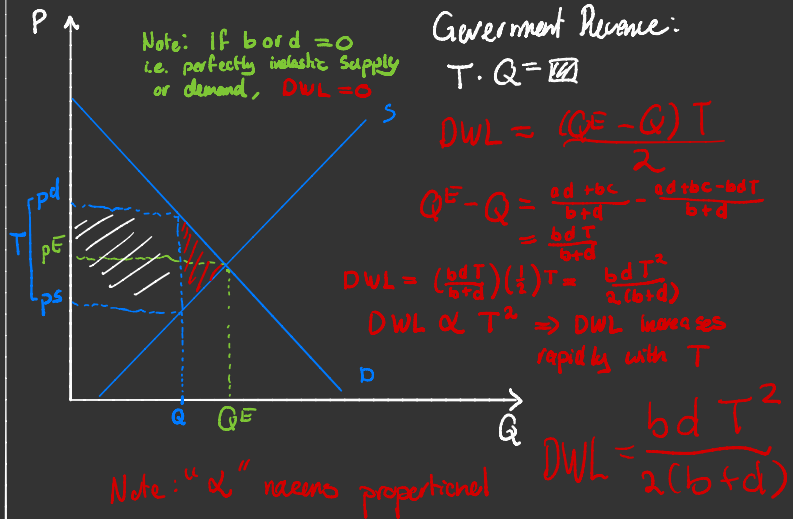
Plug-in either P^D or P^S in the respective schedule to obtain Q

$$Q = a - b \left(\frac{a - c + dT}{b + d} \right) = c + d \left(\frac{a - c - bT}{b + d} \right)$$

$$Q = \frac{ad + bc - bdT}{b + d} \quad \text{Eq. } Q \text{ after tax}$$

Note that if $T = 0$, we obtain the unrestricted equilibrium:

$$P^E = \frac{a - c}{b + d}; \quad Q^E = \frac{ad + bc}{b + d}$$



Tax Incidence:

$$p^E = \frac{a-c}{b+d}$$

Incidence

Consumers:

$$p^d = \frac{a-c+dT}{b+d}$$

$$\begin{aligned} \frac{p^d - p^E}{T} &= \frac{\frac{a-c+dT}{b+d} - \frac{a-c}{b+d}}{T} \\ &= \frac{\frac{dT}{b+d}}{T} = \frac{d}{b+d} \end{aligned}$$

Producers

$$p^s = \frac{a-c-bT}{b+d}$$

$$\begin{aligned} \frac{p^E - p^s}{T} &= \frac{\frac{a-c}{b+d} - \frac{a-c-bT}{b+d}}{T} \\ &= \frac{\frac{bT}{b+d}}{T} = \frac{b}{b+d} \end{aligned}$$

$$TI_D = \frac{d}{b+d} \cdot \frac{P/Q}{P/Q} = \frac{d P/Q}{b \frac{P}{Q} + d \frac{P}{Q}} = \frac{\epsilon_{s,p}}{|\epsilon_{d,p}| + \epsilon_{s,p}}$$

$\epsilon_{s,p} = d \frac{P}{Q}$: point elasticity of Supply
for a linear supply curve

$\epsilon_{d,p} = -b \frac{P}{Q}$: point elasticity of demand
for a linear demand curve

$$|\epsilon_{d,p}| = b \frac{P}{Q}$$

$$TI_S = \frac{b}{b+d} \cdot \frac{P/Q}{P/Q} = \frac{|\epsilon_{d,p}|}{|\epsilon_{d,p}| + \epsilon_{s,p}}$$

Tax Incidence depends on
the elasticities

Ex: $\epsilon_{s,p} = 0$, Perfectly Inelastic $TI_D = 0$

$|\epsilon_{d,p}| = 0$, Perfectly Inelastic $TI_S = 0$

$$Q_0 = 100 - 2P^d; \quad Q_s = 10 + P^s; \quad T = 10$$

$$10 = P^d - P^s \Rightarrow P^s = P^d - 10$$

$$\hookrightarrow Q_s = 10 + P^d - 10 \Rightarrow Q_s = P^d$$

$$100 - 2P^d = P^d \Rightarrow P^d = \frac{100}{3} \approx 33.33$$

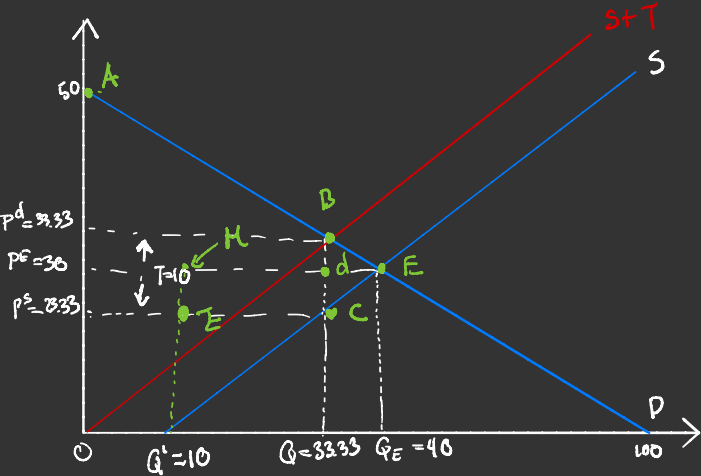
$$P^s = P^d - 10 = \frac{100}{3} - 10 = \frac{70}{3} \approx 23.33$$

$$Q = 100 - 2\left(\frac{100}{3}\right) = 10 + \frac{70}{3} = \frac{100}{3} \approx 33.33$$

If $T = 0$;

$$P^d = P^s = P^E = \frac{100 - 10}{2 + 1} = 30$$

$$Q^E = 100 - 2(30) = 10 + 30 = 40$$



Area		Amount
Tax paid by consumers	$P^d B d P^E$	$(P^d - P^E)Q = (33.33 - 30)(33.33) = 110.99$
Tax paid by producers	$P^E d C P^s$	$(P^E - P^s)Q = (30 - 23.33)(33.33) = 222.34$
Total Government Tax Collection	$P^d B C P^s$	$(P^d - P^s)Q = T \cdot Q = 10 \cdot 33.33 = 333.33$
Consumer Surplus		Producer Surplus
Before Tax	$P^E A E = \frac{1}{2}(50 - 30)(40) = 400$	$O P^E E Q^1 = \frac{1}{2}(40 - 10)(30) + (30)(10) = 750$
After Tax	$P^d A B = \frac{1}{2}(50 - 33.33)(33.33) = 277.81$	$O P^s C Q^1 = \frac{1}{2}(33.33 - 10)(23.33) + (23.33)(10) = 506.47$
Dead weight Loss		$\frac{1}{2} A P \Delta Q = \frac{1}{2} T (Q^E - Q) = \frac{1}{2}(10)(40 - 33.33) = 33.35$

$$SS^{\text{before}} = 1150$$

$$SS^{\text{after}} = 783.28$$

$$R = 333.33$$

$$DWL = 33.35$$

$$SS^{\text{before}} = SS^{\text{after}} + R + DWL$$

$$\Delta SS = SS^{\text{before}} - SS^{\text{after}} = R + DWL$$

Change in SS funds government revenue (used public goods services)
+
DWL: amount nobody appropriates!

Subsidies:

A subsidy is a negative tax ($-T$), so that

$$p^d - p^s = -T \Rightarrow p^d + T = p^s$$

If we call $S = T \Rightarrow p^d + S = p^s$

"Shifting" the supply:

$$Q_s = c + d(p^d + S)$$

"Shift" the demand:

$$p^d = p^s - S$$

$$Q_D = a - b(p^s - S)$$



Equilibrium: $p^d, p^s, Q : c + d(p^d + S) = a - b p^d$

$$p^d = \frac{a - c - dS}{b + d}$$

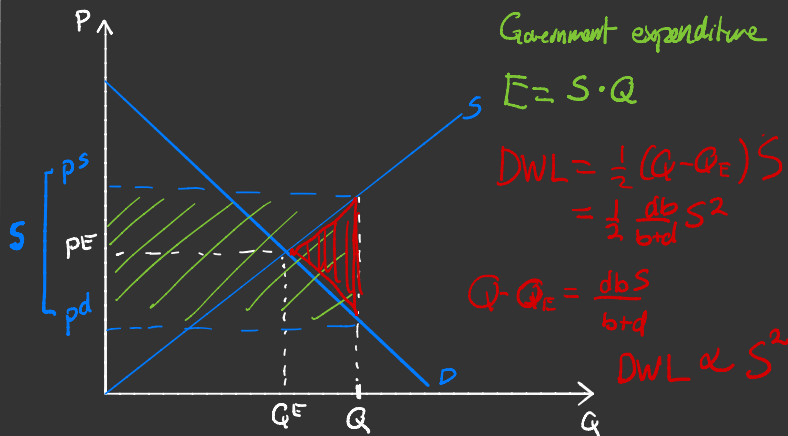
$$p^s = p^d + S = \frac{a - c - dS}{b + d} + S = \frac{a - c - dS + bS + dS}{b + d}$$

$$p^s = \frac{a - c + bS}{b + d}$$

$$Q = a - b\left(\frac{a - c - dS}{b + d}\right) = c + d\left(\frac{a - c + bS}{b + d}\right) = \frac{ad + bc + dbS}{b + d}$$

Again note that: if $S = 0$, we obtain the unrestricted equilibrium:

$$p^E = \frac{a - c}{b + d}, Q^E = \frac{ad + bc}{b + d}$$



Subsidy Incidence

Consumers $\frac{p^E - p^d}{S} = \frac{d}{b + d} \Rightarrow \frac{\epsilon_{S,p}}{|\epsilon_{D,p}| + \epsilon_{S,p}}$

Producers $\frac{p^s - p^E}{S} = \frac{b}{b + d} \Rightarrow \frac{|\epsilon_{D,p}|}{|\epsilon_{D,p}| + \epsilon_{S,p}}$

$$Q_D = 100 - 2P^D; Q_S = 10 + P^S;$$

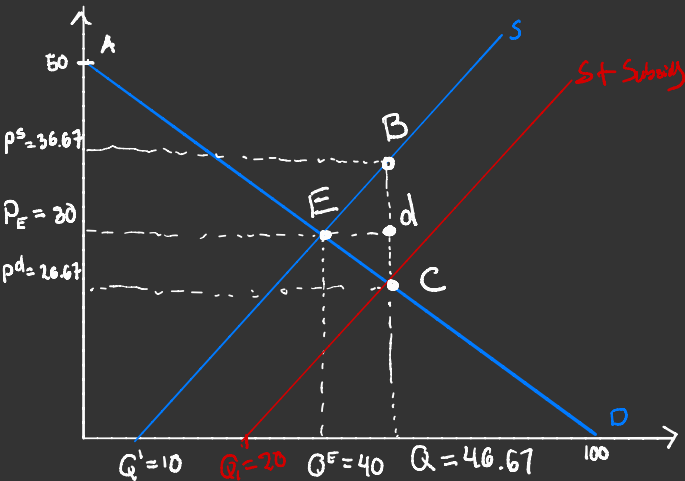
without subsidy $P^D = P^S = P^E = 30$

$$Q^E = 40$$

$$P^D = \frac{100 - 10 - 1)(10)}{2+1} = 26.67$$

$$P^S = \frac{100 - 10 + 2(10)}{2+1} = 36.67$$

$$Q = 100 - 2(26.67) = 10 + 36.67 = 46.67$$



	Area	Amount
Subsidy received by consumers	$P^E d c P^D$	$(P^E - P^D)Q = (30 - 26.67)(46.67) = 155.411$
Subsidy received by producers	$P^S B d P^E$	$(P^S - P^E)Q = (36.67 - 30)(46.67) = 311.29$
Total Government Subsidy	$P^S B C P^D$	$(P^S - P^D)Q = (36.67 - 26.67)(46.67) = 466.7$

	Consumer Surplus	Producer Surplus
Before Subsidy	$P^E A E = \frac{1}{2}(50 - 30)(40) = 400$	$O G' E P^E = (10)(30) + \frac{1}{2}(30)(40 - 10) = 750$
After Subsidy	$P^D A C = \frac{1}{2}(50 - 26.67)(46.67) = 544.46$	$O Q' B P^S = (10)(36.67) + \frac{1}{2}(36.67)(46.67 - 10) = 1039.04$

DWL $B E C$

$$\frac{1}{2} \Delta P \Delta Q = \frac{1}{2} S \Delta Q = \frac{1}{2}(10)(46.67 - 40) = 33.335$$

$$\left. \begin{array}{l} SS^{\text{before}} = 1150 \\ SS^{\text{after}} = 1583.5 \\ E = 466.7 \\ DWL = 33.35 \end{array} \right\} \begin{array}{l} SS^{\text{after}} + DWL = SS^{\text{before}} + E \\ \Delta SS + DWL = E \end{array}$$

Gov. Expenditure funds ΔSS
but some of it is lost because nobody appropriates it!